

Heart of Betrayal: A PIN Inference Attack Leveraging Photoplethysmography on Wearables

Presented by: Ishanya

Roll Number: 21329

Authors: Shiyue Huang*, Yuchen Su*, Hongbo Liu*, Huan Dai†, Bo Liu‡

* *University of Electronic Science and Technology of China, China*

† *Suzhou University of Science and Technology, China*

‡ *Auburn University, USA*

Conference: *IEEE Proceedings of the 2024 27th International
Conference on Computer Supported Cooperative Work in Design*

BACKGROUND

Background and Objectives

Background:

- PPG sensors measure blood circulation variations by illuminating the skin and tissue with LEDs and measuring the reflected light. They are non-invasive and low-cost.
- Motion induces stronger PPG changes, which can leak sensitive information like PINs which can be used for security breaches.

Objectives:

- PPGLogger: a side-channel attack using PPG sensors for PIN inference.
- Milli-meter level accuracy for action recovery.
- Minimize noise and detect keystrokes with high accuracy.
- Zero-shot or few shot inference.*

METHODOLOGY

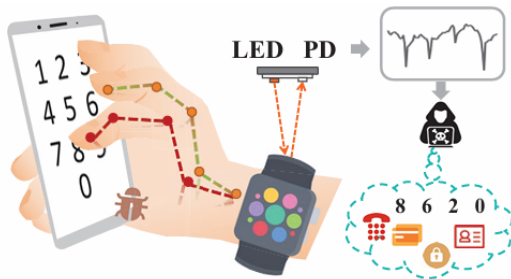


Fig. 1: PIN inference based on PPG measurements.

Methodology: Experiment Design

■ Devices:

- OPPO Watch Pro 3 smartwatch on right hand
 - Connected to laptop via WiFi.
 - 11 channels of LEDs (red, infrared, and green) available.
- Xiaomi 12 phone for keystroke logging.
- A malware program on the smartwatch collects PPG measurements all the time and labelled keystroke signals via CAPTCHA prompts and login.
- Goal: Infer the victim's PIN from the smartwatch data.

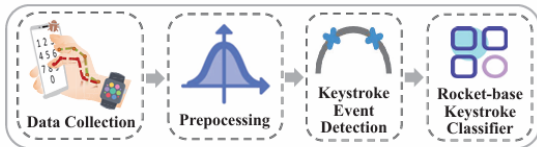


Fig. 4: Workflow of PPGLLogger.

Methodology: Data Collection

Participants:

- 10 participants (3 females, 7 males).

Data

- PPG measurements continuously recorded by OWP3.
- Data acquired at a maximum 50Hz sampling rate from 9 PPG channels (red, infrared, green LEDs).
- Each participant performed 50 trials of 4-digit PIN entry from the RockYou database.
- Total 5000 key-strokes divided into 20% data labeled for training, rest for testing.
- Real-time data acquisition configured in `SENSOR_DELAY_FASTEST` mode.

Feasibility Study: Single Stroke Analysis

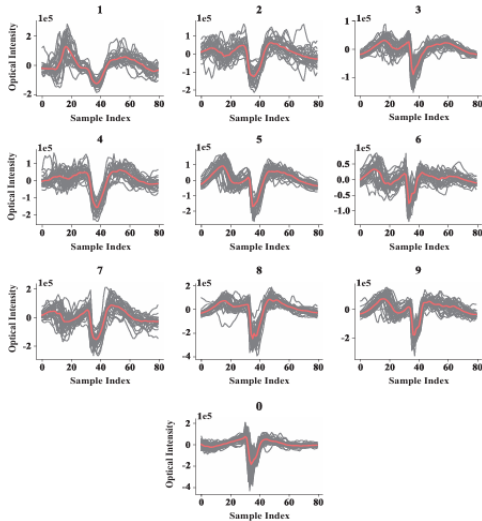
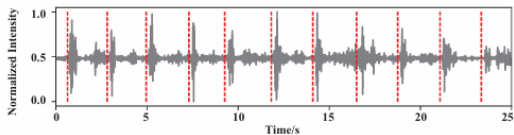


Fig. 2: PPG measurements for different keystrokes.

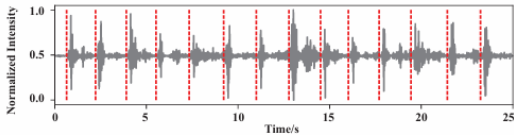
Feasibility Study: Inter-Key Distance Analysis

- Keystrokes with greater inter-key distances (IKD) require more repositioning time.
- IKD categorized as:
 - Short ($IKD \in [0, \sqrt{2}]$)
 - Medium ($IKD \in [\sqrt{2}, \sqrt{5}]$)
 - Long ($IKD \in [\sqrt{5}, \sqrt{10}]$)
- Experiment (25s):
 - Short-range: 24 keystrokes
 - Medium-range: 14 keystrokes
 - Long-range: 11 keystrokes
- Distant keys caused larger PPG variations than closer ones.
- Horizontal key presses showed more variation than vertical.
- Longer IKD results in slower keystrokes, aiding PIN inference.

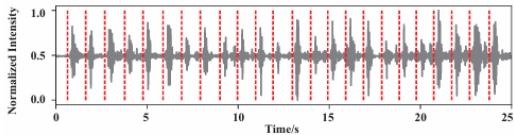
Inter-Key Distance Analysis



(a) Long-distance keystrokes



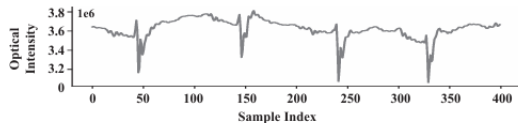
(b) Medium-distance keystrokes



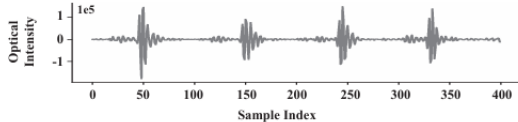
(c) Short-distance keystrokes

Fig. 3: Different distance keystrokes.

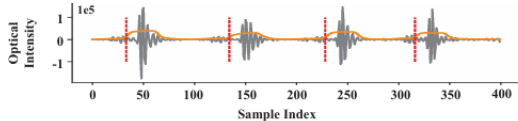
Methodology: Preprocessing



(a) Raw PPG measurement



(b) PPG measurement after high-pass filtering



(c) Segmentation

Fig. 5: Preprocessing process.

Methodology: Preprocessing

Preprocessing Steps:

■ High-pass Filtering:

- PPG signals contain physiological (heartbeat, respiration) and keystroke-induced signals.
- 2 Hz high-pass filter removes physiological signals .

■ Min-Max Normalization:

- Reduces individual variability by scaling PPG data to (0,1).
- Formula:

$$S'_i = \frac{S_i - \min(S)}{\max(S) - \min(S)}$$

Keystroke Event Detection: To extract pulse wave corresponding to single keystroke

- 10 Hz high-pass filter removes motion signals.
- Averaging filter smooths signal for stability.
- Keystroke events segmented by thresholding PPG peaks.

Rocket-based Keystroke Classifier

Keystroke Inference Model:

- **ROCKET Architecture**
- **Stochastic convolutional kernels**

Classification Model:

- **Ridge regression classifier.**
- Cross-validation.
- The optimal classification parameters are computed using the training data, and the model's effectiveness is evaluated on the test set.

Evaluation:

- Model evaluated on test data to identify target user's keystrokes.
- Top-k accuracy metrics used for evaluation ($k=1$ and $k=3$).

RESULTS AND ANALYSIS

Results: With representative subjects

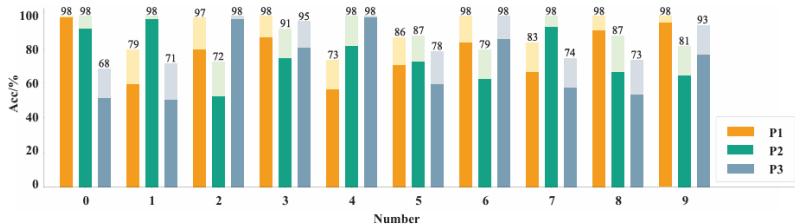


Fig. 6: Classification accuracy of single keystroke.

- Single-button classification accuracies-P1, P2, P3 subjects
- **Top1 Accuracy:** Exceeds 50% for all keys, some keys surpass 95%.
- Achieves millimeter-level action recognition.

Results: Overall Performance

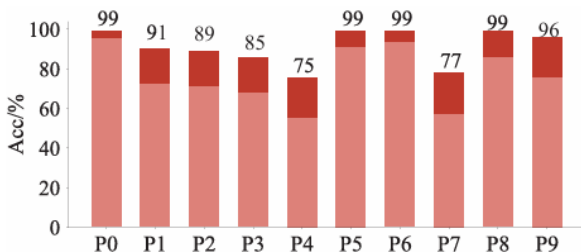


Fig. 7: Overall performance of PPGLLogger.

- **Average Top1 Accuracy:** 76.3%
- **Average Top3 Accuracy:** 91.6%
- Users who maintain stillness during keystrokes tend to achieve higher classification accuracy.

Results: Practical factors

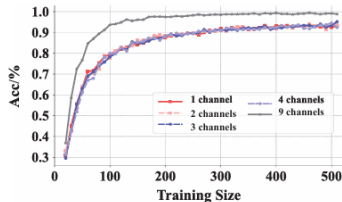


Fig. 8: Impact of training dataset size.

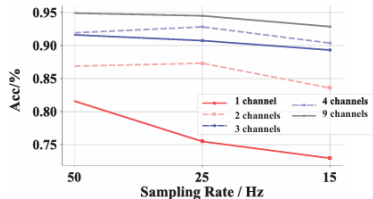


Fig. 9: Impact of sampling rate.

■ Training Dataset Size:

- Increasing training samples improves accuracy.

■ Impact of Channel Number:

- Multiple channel types improve accuracy.

■ Sampling Rate:

- Power consumption affected by sampling rate.
- Attack feasible on lower-end devices with reduced sampling rates and multiple distinct channels.

DISCUSSIONS

Limitations and Recommendations

Limitations:

- Limited sample size may affect generalizability.
- Focus on short PINs; testing with longer PINs is needed.
- Only one model employed; could be experimented with.
- Inconsistencies, in units and images.

Recommendations:

- **Few-Shot Learning:** Improve model efficiency with minimal data for personalization.
- **Data Privacy:** Ensure data never leaves the device for better user privacy.
- **Incorporate GPS Data:** Combine GPS data with PPG data.
- **Model Expansion:** Test with various models.

THANK YOU =)